

Mini Project Synopsis on

Proctored Responsive Web Application for Shree Science Academy

Submitted in partial fulfillment of the requirements for the Mini Project of

THIRD YEAR BACHELOR OF TECHNOLOGY

in

COMPUTER SCIENCE & ENGINEERING

Submitted by

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Sant Dnyaneshwar Shikshan Sanstha's
**ANNASAHEB DANGE COLLEGE OF ENGINEERING AND
TECHNOLOGY, ASHTA**
(An Empowered Autonomous Institute)

SHIVAJI UNIVERSITY, KOLHAPUR
Academic Year: 2025-26

TVS Inventory Management System

SHREE TVS , Near ST Stand rd, ISHWARPUR, Tal - Walwa, Dist -
Sangli, Maharashtra, Pin Code – 415409

Contact – 7666701030

Registration Number (RN) -

Date-

To,
Sandeep Sutar Sir and Shubhangi Patil Mam
Project Supervisor,
Department Of Computer Science & Engineering.
Annasaheb Dange College of Engineering and Technology, Ashta.

Subject- Regarding acceptance letter of sponsorship for Third Year Project of ADCET Students.

Dear Sir,

I Mr. Dhananjay Thorat, owner of SHREE TVS , Ishwarpur, Sangli. would like to inform you that, we are accepting the sponsorship for **Third Year Project assigned to students of Computer Science and Engineering Department** at Annasaheb Dange College of Engineering and Technology, Ashta.

Project Title “TVS Inventory Management System”

Project Group members,

1. Mr. Mukul Ramesh Kumbhar (URN-23031012)
2. Mr. Aditya Ananda Kumbhar (URN-23031016)
3. Mr. Anant Kirankumar Thorat (URN-23031001)
4. Mr. Pratik Chandrakant Mali (URN-23031047)

Kindly please inform you that, you should complete project on or before 30th April 2025 and deploy the developed system at our place, also you should submit the source code and project documentation in hard and softcopy. After completion of said formalities, we will release the remuneration of Rs.10,000/- (Ten Thousand Only) for this project as per our agreement.

Thanking you,


Mr. Dhananjay Thorat
Owner Shree TVS
Ishwarpur Mob:- 7666701030.



Abstract

The rapid growth of automobile dealerships and spare parts businesses has increased the need for efficient systems to manage inventory and sales operations. Many small and medium-scale dealerships still rely on manual methods such as registers or basic spreadsheets for managing spare parts inventory and billing records. These traditional approaches often lead to problems such as inaccurate stock tracking, slow billing processes, and difficulty in analyzing sales performance.

The proposed project titled “**TVS Inventory Management System**” aims to develop a web-based application that helps spare parts dealerships manage their inventory and billing processes efficiently. The system will allow users to add, update, and manage spare parts details, track stock availability, generate customer bills, and produce sales reports for better decision-making.

The system will provide a simple and user-friendly interface that enables staff to quickly search spare parts, check available stock, and generate bills without manual calculations. It will also provide report generation features that help the dealership analyze daily, monthly, or custom sales data.

The proposed system will be developed using modern web technologies and will follow a modular architecture consisting of inventory management, billing, and reporting modules. By implementing this system, the dealership can reduce manual workload, improve operational efficiency, and maintain accurate records of spare parts inventory.

Keywords: Inventory Management System, Spare Parts Management, Billing System, Web Application, Sales Reporting, Stock Monitoring.

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1 Introduction

Inventory management plays a vital role in the efficient functioning of automobile spare parts dealerships. Spare parts dealers deal with a large number of products that need to be tracked and managed properly to ensure smooth business operations. Traditional methods of maintaining inventory using registers or spreadsheets are time-consuming and prone to human errors.

With the advancement of digital technologies, businesses are gradually shifting toward automated systems that simplify operational tasks. A digital inventory management system can significantly reduce the complexity of maintaining records while improving the accuracy and efficiency of business processes.

The TVS Inventory Management System is designed to provide a digital solution for managing spare parts inventory, billing, and sales reports in a TVS dealership. The system will allow dealership staff to easily add new parts, update stock quantities, generate customer bills, and view reports related to sales and inventory.

The primary goal of this project is to create a reliable and user-friendly system that helps dealerships manage their spare parts business more efficiently and improve overall productivity.

1.1 Background and Context

In the automobile spare parts industry, maintaining accurate inventory records is essential for smooth operations. Dealerships often manage hundreds of spare parts that need to be tracked regularly. When manual systems are used, it becomes difficult to maintain accurate stock records and generate quick bills for customers.

Many modern inventory management systems exist in the market, but they are often expensive and complex, making them unsuitable for small dealerships. As a result, many dealers continue using traditional methods despite their limitations.

The proposed **TVS Inventory Management System** is intended to provide a cost-effective and easy-to-use solution specifically designed for small and medium-scale

spare parts dealerships. The system focuses on improving efficiency, reducing manual errors, and providing useful insights into sales and inventory data.

1.2 Purpose

The main purpose of the **TVS Inventory Management System** is to develop a web-based application that helps spare parts dealerships manage their inventory and billing processes efficiently.

The system is intended to:

- Maintain accurate records of spare parts inventory
- Generate customer bills quickly and accurately
- Track stock availability in real time
- Provide sales reports and analytics
- Reduce manual workload and operational errors

2 Literature Survey

Year	System / Paper	Methodology	Advantages	Limitations	Year
2020	Manual Inventory Registers	Paper-based records	Low cost	High chances of errors	2020
2021	Spreadsheet Inventory Systems	Excel-based tracking	Easy to maintain	Limited automation	2021
2022	Retail Inventory Software	Database-driven systems	Accurate tracking	Expensive for small businesses	2022
2023	Cloud Inventory Platforms	Web-based inventory tools	Remote access	Requires internet connectivity	2023

Observation:

Most existing systems either depend on manual tracking or expensive commercial software. Therefore, a simple and customizable web-based inventory management system suitable for spare parts dealerships is required.

3 Problem Statement

- **IDEAL:** TVS spare parts dealerships should have an efficient digital system to manage inventory, generate bills, and analyze sales data easily.
- **REALITY:** Many dealerships still rely on manual methods for inventory management and billing, which leads to inefficiencies and errors.
- **CONSEQUENCES:**
 - Incorrect stock records
 - Slow billing process
 - Difficulty in tracking sales performance
 - Increased possibility of inventory mismanagement
- **PROPOSAL:** Develop a proctored responsive web application with authentication, monitoring, logging, analytics, and responsive UI.
- **PROPOSAL:** To develop a **TVS Inventory Management System** that automates inventory tracking, billing, and report generation for spare parts dealerships.

4 Objectives

- To design a user-friendly web interface for managing spare parts inventory
- To develop a billing system for generating customer invoices
- To maintain accurate records of spare parts stock
- To generate sales reports for analysis and decision-making
- To improve efficiency and reduce manual errors in dealership operations

5 Scope

Included:

- Spare parts inventory management
- Customer billing system
- Sales transaction recording
- Stock availability tracking
- Sales reports and analytics
- User authentication and access control

Excluded / Limitations:

- Integration with large-scale ERP systems
- AI-based demand forecasting
- Multi-branch inventory synchronization

6 Proposed Work

6.1 Methodology (Proposed modules, techniques, algos, flowchart/architecture,)

System Architecture:

The proposed **TVS Inventory Management System** follows a simple three-layer architecture:

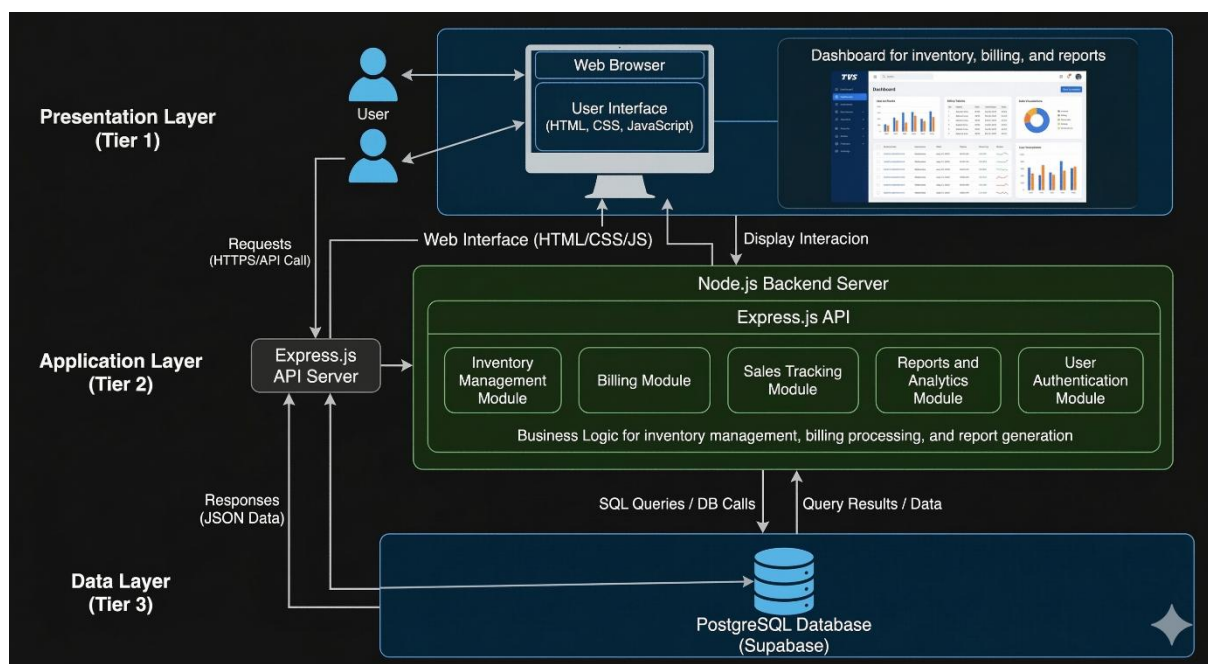
- **Presentation Layer:** Web interface developed using **HTML, CSS, and JavaScript** for interacting with the user.
- **Application Layer:** JavaScript functions that process inventory operations, billing logic, and part searching.
- **Data Layer:** Storage system that maintains spare parts data, stock quantities, and billing records.

Core Modules:

1. **Inventory Management Module:** Handles adding, updating, and deleting spare parts in the inventory while maintaining stock records.
2. **Billing Module:** Allows the user to create bills by selecting spare parts, entering customer details, and calculating total cost automatically.
3. **Part Search Module:** Enables quick searching of spare parts using part name or part number through a dynamic search bar.
4. **Reporting Module:** Generates simple reports such as available stock, sold parts, and billing summaries.

Techniques and Algorithms:

- **Dynamic Search Filtering:** Displays matching parts as the user types in the search field.
- **Automatic Stock Update:** Inventory quantity decreases automatically when a part is sold.
- **Billing Calculation:** Calculates total price based on part price and quantity.
- **Data Storage:** Maintains inventory and billing records for future reference.



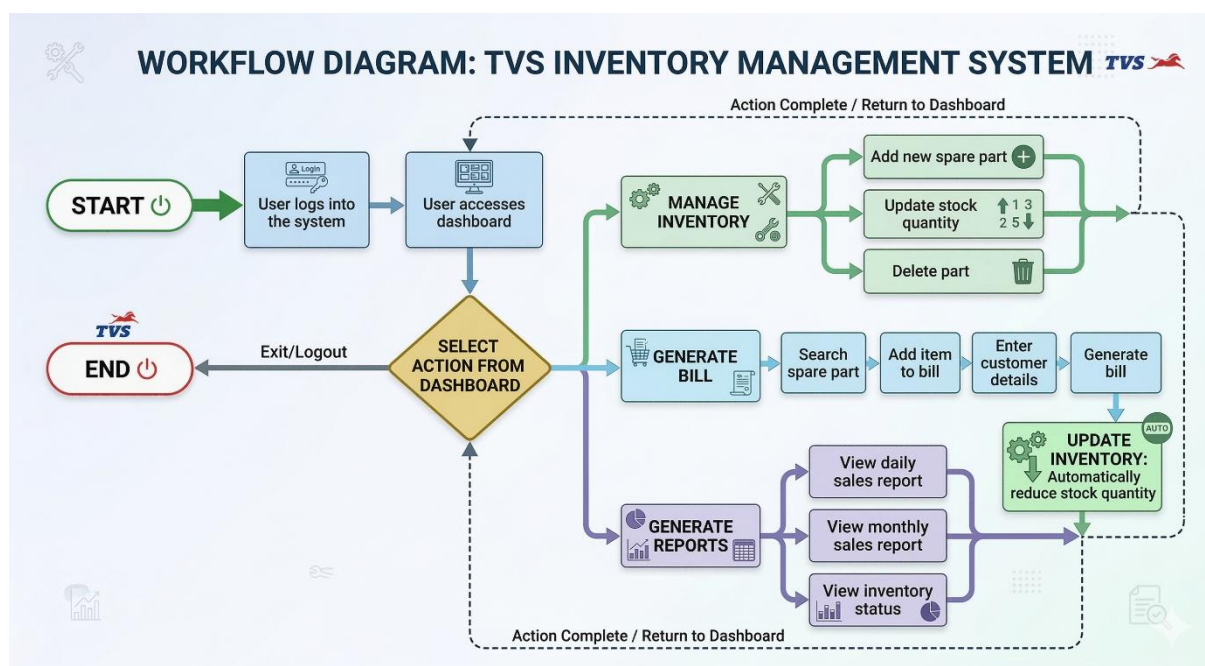
6.2 Software and Hardware requirements and availability

Software Requirements:

- **Frontend** – HTML5, CSS3, JavaScript\
- **Backend** – Node.js, Express.js
- **Database** – PostgreSQL / Supabase
- **Hosting / Deployment** – Vercel (Frontend Hosting), Render (Backend Deployment)
- **Monitoring Tool** – UptimeRobot
- **Development Tools** – Visual Studio Code / Windsurf , Git
- **Browser** – Google Chrome / Mozilla Firefox

Hardware Requirements:

- **Development System** – Personal Computer or Laptop with minimum 8GB RAM
- **Server** – Cloud-based hosting platform
- **Client Devices** – Desktop computers, laptops, or smartphones with modern web browsers



Availability:

- All required software tools are freely available or have free tiers
- Cloud hosting providers offer scalable solutions with pay-as-you-go pricing
- Standard hardware components are readily available in the market
- Open-source technologies ensure long-term availability and support

7 Expected Outcome

After successful implementation, the TVS Inventory Management System will provide the following outcomes:

- Efficient management of spare parts inventory
- Faster billing and customer service
- Accurate stock monitoring
- Easy generation of sales reports
- Improved operational efficiency for spare parts dealerships

The system will help reduce manual work and provide a reliable digital solution for managing spare parts inventory.

8 Schedule

- Literature Survey – Weeks 1–2
- Design – Weeks 3–4
- Development – Weeks 5–10
- Testing – Weeks 11–12
- Deployment – Week 13
- Documentation – Week 14

References (List of books, websites, blogs/articles and papers.)

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(An Autonomous Institute affiliated to Shivaji University, Kolhapur)

Department of Computer Science and Engineering

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Place:

Name of Guide

GUIDE

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